

MERN STACK INTERNSHIP – DAY 14 ASSIGNMENT REPORT

Company

Insta Dot Analytics

Submitted By

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FOR SOURCE CODE:

GIT HUB :- <https://github.com/chayan-soni/MERN-Internship>

Objective

The objective of this task was to implement a Wishlist feature in the MERN E-Commerce application. The task included creating CRUD APIs for Wishlist management, integrating the APIs with the React frontend, displaying notifications using Toasts, and pushing the updated code to GitHub.

Task 1: Implement Wishlist Functionality

A Wishlist feature was added to the MERN E-Commerce project to allow users to save products for future purchases. This functionality enhances the user experience by providing a convenient way to manage favorite products.

Features Implemented:

- Add products to Wishlist
- View Wishlist items
- Remove products from Wishlist
- Prevent duplicate products from being added
- Restrict Wishlist access to authenticated users

Task 2: Create CRUD APIs for Wishlist

RESTful APIs were developed using Node.js, Express.js, and MongoDB to manage Wishlist operations.

Backend Implementation:

- Created Wishlist schema/model in MongoDB.
- Developed controller functions for CRUD operations.
- Added authentication middleware for secure access.
- Implemented error handling and validation checks.

Task 3: Integrate Wishlist APIs with React Frontend

The Wishlist APIs were integrated into the React application to provide real-time interaction between the frontend and backend.

Frontend Enhancements:

- Added Wishlist button on product cards.
- Added Wishlist option on product detail pages.
- Connected API calls using Axios.
- Updated UI dynamically based on API responses.
- Managed Wishlist state effectively.

Task 4: Display Success/Error Notifications Using Toasts

Toast notifications were implemented to improve user feedback and interaction.

Notifications Added:

- Success message when a product is added to Wishlist.
- Success message when a product is removed from Wishlist.
- Error message when API requests fail.
- Validation messages for invalid operations.

Benefits:

- Better user experience.
- Instant feedback on actions.
- Improved application responsiveness.

Task 5: Push Updated Code to GitHub

After successful implementation and testing, all changes were committed and pushed to the GitHub repository.

Git Commands Used:

```
git add .  
git commit -m "Implemented Wishlist functionality with CRUD APIs and  
frontend integration"  
git push origin main
```

Testing Performed

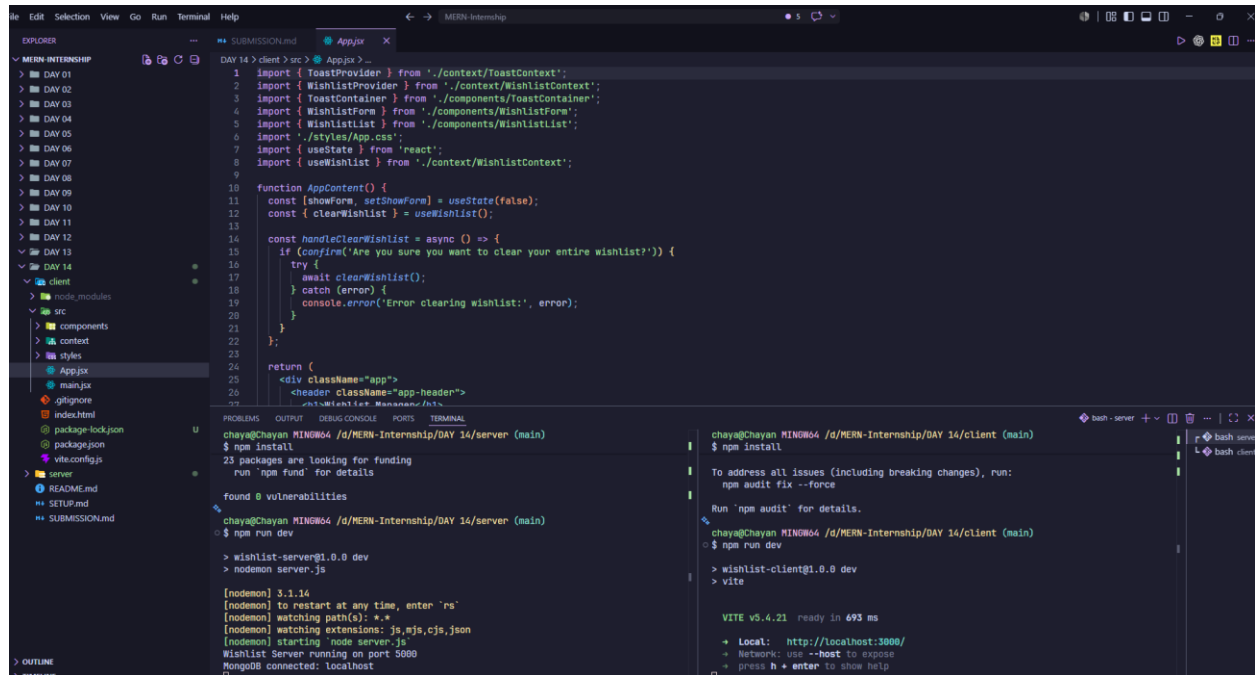
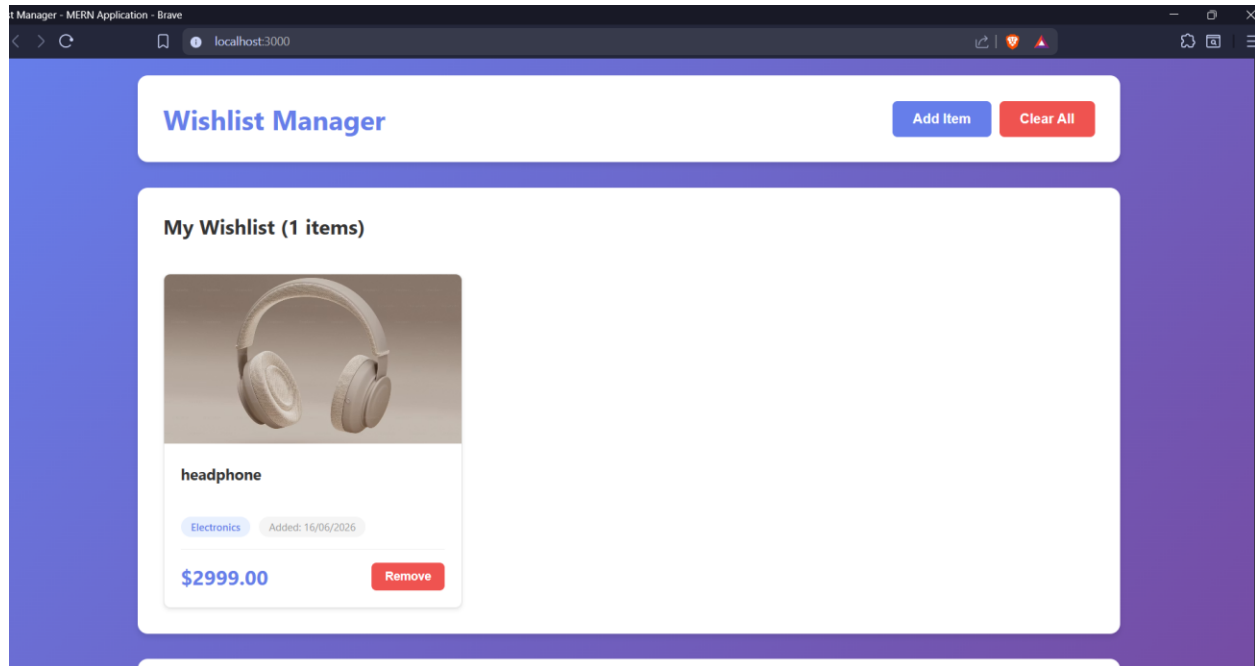
- Verified product addition to Wishlist.
- Verified Wishlist retrieval functionality.
- Tested product removal from Wishlist.
- Checked API responses and error handling.
- Validated Toast notifications.
- Confirmed successful integration between frontend and backend.

Outcome

Successfully implemented the Wishlist feature in the MERN E-Commerce project. The application now supports complete Wishlist management through CRUD APIs, React frontend integration, Toast notifications, and version control using GitHub. The feature was tested thoroughly and is functioning as expected.

Conclusion

This task provided practical experience in full-stack development using the MERN stack. It strengthened understanding of REST API development, MongoDB data management, React integration, state management, user feedback mechanisms, and GitHub workflow. The Wishlist functionality significantly enhances the usability and overall user experience of the E-Commerce application.



Add Item to Wishlist

Product ID *

Product Name *

Price *

Image URL

Category



Add to Wishlist